Strong Readouts, Local Levers: A Steering Audit of Gemma-3-4B-IT

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Abstract

Predictive internal signals are often treated as natural targets for steering language models, but their reliability as intervention handles remains unclear. We test this assumption in Gemma-3-4B-IT. In a cross-family FaithEval comparison, sparse autoencoder (SAE) features and magnitude-ranked H-neurons achieve similar held-out detection quality (AUROC 0.848 vs. 0.843), yet only H-neurons produce a reliable compliance dose-response (+2.1 pp/α [1.4, 2.8]; neuron-minus-SAE gap excludes zero). When steering works, it remains surface-local: ITI improves TruthfulQA answer selection but reduces TriviaQA bridge accuracy, most often through wrong-entity substitution. Measurement choices also change the conclusion: truncation and scoring granularity alter the apparent jailbreak result, and a held-out evaluator audit reveals rubric-dependent disagreement. We organize these findings as a four-stage audit framework (measurement, localization, control, and externality) and argue that strong readouts are insufficient evidence for good steering targets.

1. Introduction

A predictive internal signal (a neuron, feature, or direction that reliably discriminates between behavioral categories on held-out data) is a tempting steering target. If a feature predicts whether a model will hallucinate or comply harmfully, it is natural to expect that amplifying or suppressing it will steer behavior. This heuristic underlies much recent work: identify a strong readout, then intervene through it (Li et al., 2023; Gao et al., 2025; Arditi et al., 2024).

The heuristic sometimes works. Refusal-mediating directions produce reliable refusal modulation (Arditi et al., 2024), and hallucination-associated neurons can shift over-compliance (Gao et al., 2025). But the heuristic also sometimes fails, and the failure modes have received less systematic attention than the successes.

This paper tests the readout-to-steering heuristic empirically in Gemma-3-4B-IT (Google DeepMind, 2025). We find repeated dissociations between four stages that are routinely conflated:

1. **Measurement.** Can we trust the evaluation? Truncation and scoring granularity changed the scientific conclusion, while the evaluator audit exposed residual rubric dependence (§5).
2. **Localization.** Does the readout identify causally relevant components? SAE features matched H-neurons on detection quality but did not translate into useful control (§3).
3. **Control.** Does intervention produce the intended behavioral change? Effects were narrow and task-local (§4).
4. **Externality.** Does the effect transfer without causing harm? Wrong-entity substitution was the most common manually coded failure mode on a held-out bridge benchmark (§4).

Contributions. (1) A FaithEval comparison showing that similar held-out readout quality yields sharply different steering outcomes. (2) An externality analysis showing that answer-selection gains do not transfer to generation, with wrong-entity substitution as the most common manually coded failure mode. (3) A four-stage audit framework for evaluating mechanistic intervention claims without collapsing measurement, localization, control, and externality into one inference.

Study orientation. This paper is a single-model comparative case study in Gemma-3-4B-IT (Google DeepMind, 2025). We compare multiple intervention families (neuron scaling, SAE feature steering, and inference-time intervention via attention heads) across contextual faithfulness (FaithEval), answer selection (TruthfulQA), open-ended factual generation (TriviaQA, SimpleQA), domain

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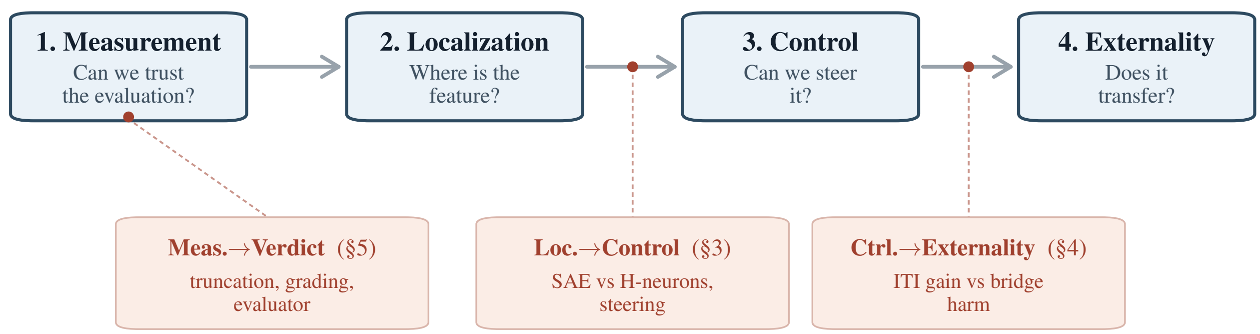

Figure 1. The Four-Stage Interpretability Scaffold and three empirical gates. Stages run left to right; each transition is a question the next stage assumes has been answered. Three gates mark where the readout-to-steering heuristic broke in our case study: one *within-stage* (Measurement→Verdict, §5: truncation and grading shifted the verdict, while the evaluator audit exposed residual rubric dependence) and two *inter-stage* (Localization→Control, §3; Control→Externality, §4). Within-stage gates tether to a stage box; inter-stage gates tether to the transition arrow.

QA (BioASQ), and jailbreak settings (JailbreakBench). All claims are surface-specific: FaithEval is a context-faithfulness / anti-compliance surface, TruthfulQA MC constrained answer selection, TriviaQA bridge and SimpleQA open-ended factual generation, and JailbreakBench a harmfulness-evaluation surface. Within that scope, the paper asks three bounded empirical questions: (1) Is held-out readout quality enough to select steering targets? (§3) (2) Do answer-selection gains transfer to open-ended factual generation? (§4) (3) Do plausible measurement choices leave the jailbreak verdict stable? (§5) These are surface-specific questions, not claims of a new steering method or a general failure of detector-based targets.

2. Related Work

Foundational probe critiques establish that high readout accuracy can arise for reasons the model does not functionally rely on (Hewitt & Liang, 2019; Elazar et al., 2021; Kumar et al., 2022). More recent work moves to the intervention question directly: within SAE feature selection, high-scoring features need not steer well (Arad et al., 2025); AxBench reports that detection and steering need not be jointly optimized (Wu et al., 2025); Wang et al. (2026) report only a weak positive association between SAE interpretability and steering utility across 90 SAEs (Kendall's $\tau_b \approx 0.298$); and Bhalla et al. (2024) describe a predict/control discrepancy for interpretability-based interventions. Li et al. (2023) already showed that ITI's multiple-choice gains can diverge from generative gains, and Opiełka et al. (2026) demonstrated that causally effective function vectors can be format-specific. Hase et al. (2023) showed that better localization does not reliably predict better editing, providing a close precedent for the localization/control dissociation. On measurement, StrongREJECT (Souly et al., 2024) showed that binary scoring can overestimate jailbreak effectiveness, and Chen & Goldfarb-Tarrant (2025) showed that evaluator artifacts materially affect safety judgments.

Concurrent SAE-centric analyses (Arad et al., 2025; Wang et al., 2026) and synthetic-benchmark evaluations (Wu et al., 2025) make detection/control divergence increasingly clear within single-family settings. The core contribution here is a cross-representational FaithEval comparison: neurons and SAE features are evaluated on the same real behavioral surface with closely matched held-out readout quality. Two supporting additions broaden that case study. First, on TriviaQA bridge generation we provide a frozen-protocol behavioral diagnosis of the transfer failure, where wrong-entity substitution is the most common manually coded right-to-wrong pattern rather than simple suppression. Second, we show that truncation and scoring granularity can change the verdict for the same representation-engineering intervention outputs, while a held-out evaluator audit reveals residual rubric dependence. We synthesize these observations in a four-stage audit framework (measurement, localization, control, externality).

3. Similar Readout Quality Does Not Guarantee Control

3.1. The Readouts Are Real

The intervention targets were selected through held-out predictive readouts. A sparse H-neuron probe following Gao et al. (2025) identified 38 positive-weight neurons out of 348,160 feed-forward neurons and achieved AUROC 0.843 (accuracy 76.5%, $n_{\text{test}} = 780$). A parallel L1 probe on Gemma Scope 2 SAE activations selected 266 positive-weight features across 10 layers and achieved AUROC 0.848 (accuracy 77.2%, $n_{\text{test}} = 782$). These are real held-out signals, though individual-weight interpretation remains fragile (Appendix A).

3.2. Comparable Readouts, Divergent Control

Both methods achieve similar held-out detection quality, and we evaluate both interventions on the same 1,000 FaithEval items (Ming et al., 2025). The 38 probe-selected neurons were scaled multiplicatively across $\alpha \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$, with $\alpha = 1.0$ as the no-op baseline. The 266 SAE features were scaled through an encode-modify-decode intervention on the same grid.

The neuron intervention showed a clear dose-response: compliance slope $+2.09$ pp/α $[1.38, 2.83]$, with perfect Spearman monotonicity ($\rho = 1.0$). Relative to the no-op baseline, compliance at $\alpha = 3.0$ increased by $+4.5$ pp $[2.9, 6.1]$. Across five unconstrained and three layer-matched random-neuron sets, all eight random seeds produced null slopes (maximum $+0.21$ pp/α); every paired neuron-minus-random difference excluded zero.

The SAE intervention did not produce the same behavior. The full-replacement H-feature slope was $+0.16$ pp/α $[-0.51, 0.84]$, with no monotonic trend ($\rho = 0.18$). A delta-only architecture that cancels reconstruction error confirmed the null: delta-only H-feature slope $+0.12$ pp/α, delta-only random-feature slope -0.09 pp/α, both indistinguishable from zero. The neuron baseline on the same three-alpha subset remained $+2.12$ pp/α.

The paired neuron-minus-SAE slope difference was $+1.93$ pp/α $[+0.94, +2.92]$ (permutation $p < 0.001$). On the same 1,000 FaithEval items, similar held-out readout quality did not predict intervention utility: H-neurons and SAE features showed comparable detector performance, but only the neuron intervention produced a robust behavioral dose-response.

3.3. Synthesis

H-neurons *did* steer FaithEval compliance, and specificity was confirmed against eight matched random controls. The refusal-direction literature (Arditi et al., 2024) is another positive case. The narrower claim is that readout quality alone is an unreliable heuristic for predicting *when* detection-based targets will steer behavior. SAE features matched H-neurons on the quantity most commonly used to justify steering (held-out AUROC) and failed on the downstream behavioral test.

A natural objection is that this comparable-readout design confounds representational basis with readout quality: SAE features and neurons differ in operator form, layer coverage, and feature granularity. That is true, and it means the clean inference is narrower than “only readout quality matters.” The claim is instead that readout quality *alone* did not suffice to predict steering utility in this comparison. We discuss this limitation further in §6.

4. Control Is Surface-Local and Can Externalize

4.1. Positive Results Are Task-Local

Before testing transfer, we establish that detector-selected interventions can steer behavior on their primary surfaces. H-neuron scaling (38 magnitude-ranked neurons) produces clear, dose-dependent effects on compliance-adjacent surfaces. On FaithEval, compliance increases by $+4.5$ pp $[2.9, 6.1]$ above the no-op baseline at $\alpha = 3.0$ (slope $+2.09$ pp/α), confirmed by 8-seed negative controls showing flat random-neuron baselines (mean slope $+0.02$ pp/α). On FalseQA ($n = 687$), the same neurons show a dose-response slope of $+1.62$ pp/α $[0.52, 2.74]$.

However, the same 38 neurons show no robust net accuracy effect on BioASQ factoid QA: -0.06 pp $[-1.5, 1.4]$ on a well-powered sample (n = 1,600, MDE $\sim$2 pp), despite substantially perturbing answer style in 1,339 of 1,600 responses. The flat accuracy endpoint coexists with substantial output perturbation, suggesting that behavioral activity and alias-accuracy movement come apart on this surface. The intervention modulates compliance-adjacent behavior but does not improve domain-specific factual accuracy under the alias metric.

4.2. ITI: Answer Selection vs. Open-Ended Generation

ITI with TruthfulQA-derived truthfulness directions (Li et al., 2023) improves answer selection. At $\alpha = 8.0$, it yields $+6.3$ pp MC1 $[3.7, 8.9]$ and $+7.49$ pp MC2 $[5.28, 9.82]$ on held-out TruthfulQA folds, with 61 incorrect→correct flips against 20 correct→incorrect flips.

This MC/generation divergence replicates the finding of Li et al. (2023) in LLaMA/Alpaca; here, SimpleQA serves as a supporting stress test, while the locked bridge benchmark below is the main externality result. The gain does not transfer to open-ended factual generation. On SimpleQA (n = 1,000) with a prompt that removes the explicit escape hatch, ITI at $\alpha = 8.0$ reduces correct-answer rate from 4.6% $[3.5, 6.1]$ to 2.8% $[1.9, 4.0]$ (paired $\Delta = -1.8$ pp $[-3.1, -0.6]$). The attempt rate collapses from 99.7% to 67.0%, while precision among attempted answers remains stable ($\sim$4%), suggesting epistemic caution rather than factual improvement. Answer-selection success is therefore not evidence for generation-level truthfulness improvement, consistent with Pres et al. (2024) and Opiełka et al. (2026).

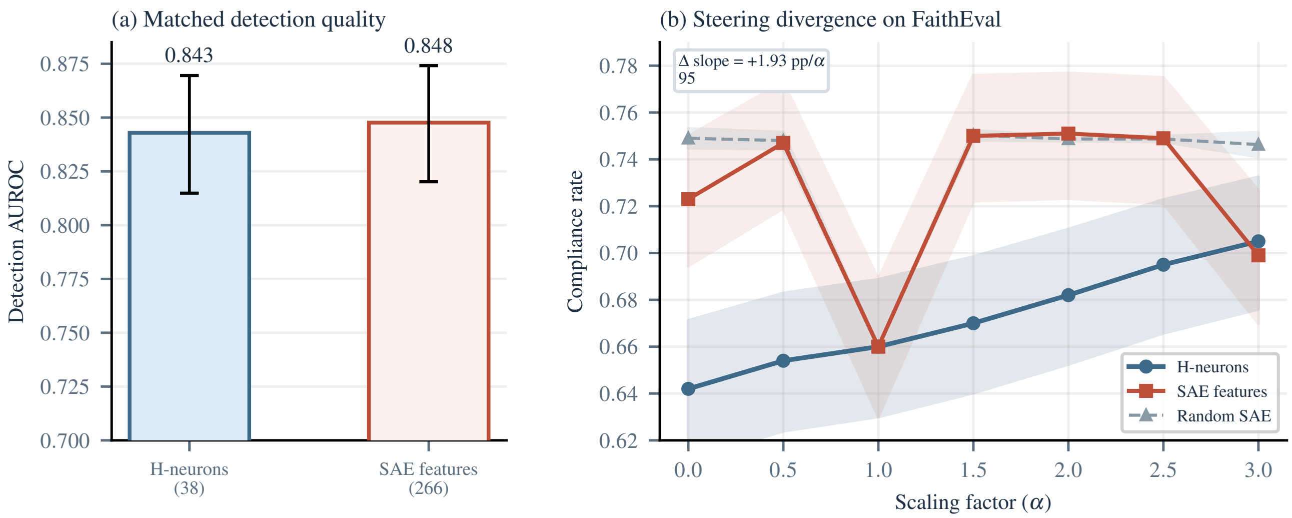

Figure 2. The anchor result: FaithEval comparable readouts, divergent control. (a) H-neurons and SAE features have similar held-out detection AUROC. (b) Only H-neurons produce a dose-dependent compliance increase; SAE features and random SAE features are both flat. The neuron-minus-SAE slope difference ($+1.93$ pp/α, 95% CI excludes zero) confirms the dissociation.

4.3. Bridge: Wrong-Entity Substitution

The TriviaQA bridge benchmark (held-out test set, $n = 500$, baseline adjudicated accuracy 45.0%) provides the most informative externality result because it reveals a specific behavioral failure mode, not just a score drop. The test set was evaluated under a one-shot frozen protocol: all parameters were locked from Phase 2 dev results, and no parameters were tuned on test data.

The intervention is active, not simply suppressive. At $\alpha = 8.0$, the baseline TruthfulQA-sourced ITI variant reduces adjudicated accuracy by -5.8 pp $[-8.8, -3.0]$ (CI excludes zero), with 43 right-to-wrong flips against 14 wrong-to-right rescues (net -29, McNemar $p = 0.0002$). Both the primary (adjudicated) and floor (deterministic) accuracy deltas exclude zero, ruling out a grading artifact.

The observed harm is not primarily refusal: NOT_ATTEMPTED counts rise from 2 to 8, a small 1.2 pp effect. Formal refusal accounts for only 2 of 43 R→W flips (5%). The dominant damage is factual corruption, not silence.

Wrong-entity substitution is the most common manually coded failure mode. Of the 43 R→W flips, 30 were manually classified as substitutions where the model replaces a correct factual answer with a different entity from the same semantic neighborhood (∼70%; Table 1). Three additional failure modes account for the remaining 13 flips: evasion/factual denial (8/43, 19%), answer dilution from verbosity (3/43, 7%), and formal refusal (2/43, 5%). This pattern is consistent with a coarse behavior-level reweighting toward nearby but wrong candidates rather than simple suppression of generation. We treat wrong-entity substitution as a behavioral diagnosis rather than a claim about the exact internal circuit mechanism.

The 14 wrong-to-right rescues show that the intervention is behaviorally active in both directions: on some questions it moves a wrong baseline answer to the correct entity, though on this surface the damage rate is larger than the rescue rate.

5. Measurement Choices Changed the Conclusion

The preceding sections established that detection quality does not predict steerability (§3) and that successful steering is narrow in scope (§4). Both conclusions rest on behavioral measurements (jailbreak harmfulness rates, severity scores, and generation-surface accuracy) that are themselves products of evaluation choices. In this section we show that those choices are part of the scientific result: truncation and scoring granularity shifted what we would have concluded about whether a given intervention worked on the same outputs, while the evaluator audit exposes residual rubric dependence on a held-out set. The general point that evaluator choices matter is established (§2); the specific finding here is that same-output measurement choices can reverse the conclusion about a representation-engineering intervention, even when two held-out evaluators tie in aggregate accuracy.

We organize the case study around the H-neuron jailbreak scaling experiment (38 probe-selected neurons, $\alpha \in \{0, 1, 1.5, 3\}$, $n = 500$ per condition), because its moderate effect size makes it sensitive to every measurement decision we examine.

Truncation hides downstream content. Short generation

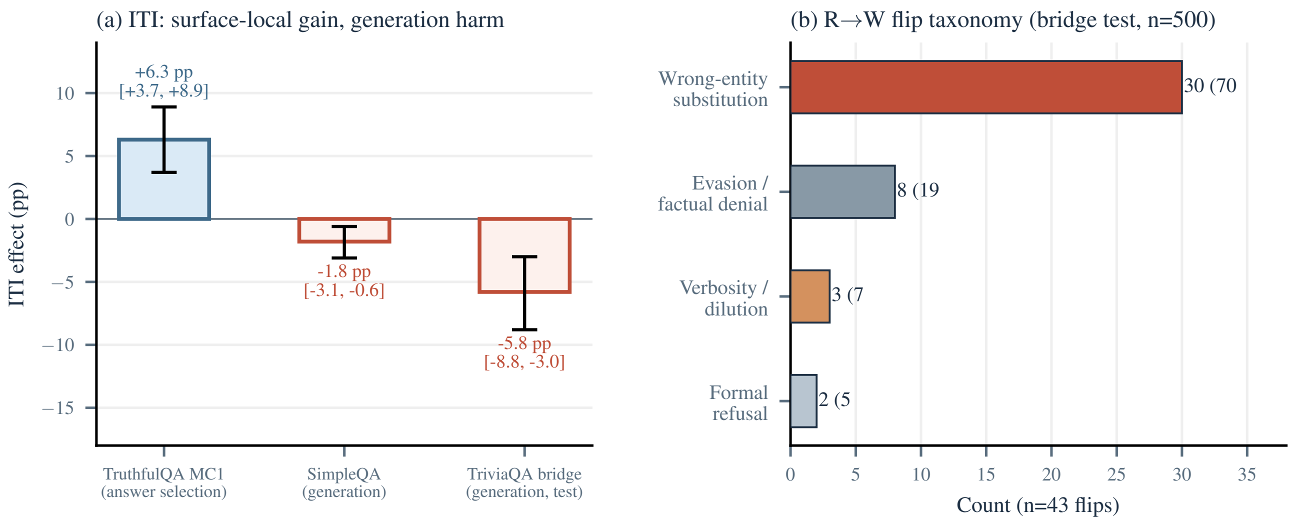

Figure 3. Surface-local control and bridge failure modes. (a) ITI improves TruthfulQA MC1 but harms both generation surfaces (CIs exclude zero). (b) Among the 43 right-to-wrong flips on the TriviaQA bridge test set, wrong-entity substitution is the most common manually coded failure mode (∼70%), not refusal.

caps (e.g., 256 tokens) can systematically bias intervention evaluations when steering changes response structure. Gemma-3-4B-IT often begins jailbreak responses with a refusal preamble and only later emits substantive harmful content. Short caps therefore preferentially capture the refusal-looking prefix and hide the downstream payload, overestimating intervention effectiveness. Full-generation scoring is therefore required for the kinds of jailbreak claims made in this paper.

Scoring granularity changes the verdict. The same H-neuron jailbreak outputs look different under binary and graded evaluation. A GPT-4o binary harmful/safe judge shows a +3.0 pp endpoint shift ($\alpha = 0 \rightarrow 3$, 152/500 → 167/500), with a confidence interval that includes zero. Under binary evaluation alone, the intervention looks weak and non-decisive.

A graded harmfulness rubric (CSV-v2) applied to the same outputs recovers a different result. The H-neuron strict harmfulness slope is $+2.30$ pp/α $[+0.99, +3.58]$, while a matched random-neuron control is flat at -0.47 pp/α $[-1.42, +0.47]$. The slope difference is $+2.77$ pp/α $[+1.17, +4.42]$ with permutation $p = 0.013$. Binary scoring washes out this signal because the effect largely lives in the movement of borderline cases into clearer refusal or clearer compliance.

This is the section's core granularity result: binary evaluation suggests a weak or null effect; graded evaluation shows a positive dose-response.

Evaluator dependence is part of the result. On a contamination-free 50-record holdout, four GPT-4o-based evaluators yield the accuracies in Table 2. No pairwise difference reaches statistical significance at $n = 50$. CSV-v3 and StrongREJECT-GPT-4o both reach 96.0% holdout accuracy with identical error sets. We retain CSV-v3 for its richer outcome taxonomy, not for accuracy superiority.

The residual disagreement is informative because it clusters on a recognizable response type: refuse-then-comply outputs. StrongREJECT's formula zeroes the final score when `refused = 1`, even if later text contains specific harmful substance. We therefore treat evaluator disagreement as construct pluralism rather than noise that can be solved by picking one preferred judge (Eiras et al., 2025).

6. A Four-Stage Audit Framework

We distill the case study evidence into a practical audit for evaluating mechanistic intervention claims. The audit has four stages: measurement, localization, control, and externality. They are not a taxonomy. They are an empirical observation about where intervention claims break in practice:

- **Measurement→Verdict.** Before deciding whether an intervention worked, one must trust the evaluation that defines the behavioral surface. In our jailbreak setting, truncation and binary-versus-graded scoring altered the conclusion about the same outputs, while the evaluator holdout exposed residual construct mismatch (§5).
- **Localization→Control.** A feature that predicts behavior on held-out data need not causally control that behavior when perturbed. SAE features matched H-neurons on detection quality, yet did not translate into useful control (§3).

Table 1. Wrong-entity substitution examples from the TriviaQA bridge test set (E0 $\alpha = 8$). The model swaps correct entities for semantically nearby incorrect ones.

Question	Baseline	ITI $\alpha = 8$
Danny Boyle's 1996 "Choose Life" film?	Trainspotting	Slumdog M. (same director)
Other musician with Buddy Holly and the Big Bopper in the Feb. 1959 crash?	Ritchie Valens	J.P. Richardson (same crash)
Family Guy character who moved to Stoolbend, Virginia?	Cleveland Brown	Peter Griffin (same show)
Dickens novel with Merdle, Sparkler, and Tattycoram?	Little Dorrit	Bleak House (same author)
Comic whose June 1938 issue introduced Superman?	Action Comics	Detective Comics (same publisher)

Table 2. Holdout evaluator accuracy (contamination-free split, $n = 50$). Prompt-clustered bootstrap 95% CIs.

Evaluator	Accuracy	95% CI
CSV-v3 (GPT-4o)	96.0%	[90.0, 100.0]
StrongREJECT (GPT-4o)	96.0%	[90.0, 100.0]
CSV-v2 (GPT-4o)	92.0%	[84.3, 98.0]
Binary judge (GPT-4o)	90.0%	[80.0, 98.0]

- **Control→Externality.** An intervention that succeeds on one surface may fail or cause harm on a nearby surface. ITI improved TruthfulQA MC1 but reduced open-ended factual accuracy, where wrong-entity substitution was the most common manually coded failure mode (§4).

Each stage transition is a distinct empirical claim. Passing one does not license claims about the next. Table 3 summarizes the minimum evidence required at each gate.

In our reading of the recent steering literature, target-surface control is reported more often than matched negative controls, cross-surface tests, or evaluator audits. In our case study, each omitted gate would have licensed a claim that the full audit rejects: without measurement audits, the jailbreak intervention looks null; without externality checks, the ITI intervention looks beneficial. The checklist is an empirical codification of the points where our own conclusions changed.

Table 3. Minimum audit for a credible steering claim. A result that clears only one or two gates is a monitoring or localization result, not a steering result.

Gate	Evidence required
Measurement	Pre-specified metric; full-generation scoring; evaluator agreement check if intervention alters output style
Localization	Held-out readout quality; but also: does steering through these components change behavior?
Control	Dose-response on target surface; matched negative control (random component, random direction); effect size with CI
Externality	At least one non-target surface; capability check; report harms and scope conditions explicitly

6.1. Five Recommendations

R1: Do not treat held-out readout quality as sufficient target-selection evidence. Metrics such as AUROC, accuracy, and probe coefficient magnitude are insufficient on their own for identifying useful steering targets. Our comparison in §3 shows that similar held-out AUROC can fail to predict intervention utility.

R2: Validate on the behavioral surface you actually care about. Answer-selection benchmarks and open-ended generation benchmarks measure different constructs, and intervention effects do not transfer reliably between them (§4).

R3: Use matched negative controls. When component search is large, post-hoc selection creates a multiple-comparisons problem. Multi-seed random controls are necessary to establish that an effect is component-specific rather than a generic perturbation artifact.

R4: Treat evaluator disagreement as information. When an intervention changes the surface form of outputs, different evaluators may reach different conclusions. This is not noise; it reflects construct mismatch (§5).

R5: Report externality costs as first-class outcomes. Interventions that help on one metric can harm others. Our bridge results show that the harm can be a specific, interpretable failure mode (§4). Cross-surface effects belong next to the headline result, not in the footnotes.

6.2. Limitations

All experiments use a single model (Gemma-3-4B-IT); the dissociations may not replicate across architectures or scales, though Gao et al. (2025) report qualitatively similar H-neuron compliance effects across six models. The comparable-readout design keeps held-out AUROC close but cannot rule out that some other property of the feature family (operator form, layer coverage) explains the steering divergence. SAE layer coverage is partial; a different SAE width or layer selection could yield non-null results. Bridge failure-mode coding is single-rater without inter-rater reliability, so exact shares are approximate. Jailbreak H-neuron specificity is single-seed ($p = 0.013$); additional seeds are needed. Concurrent work (Wang et al., 2026; Arad et al., 2025; Wu et al., 2025) already argues for interpretability/utility divergence from SAE-centric and benchmark-driven perspectives; we do not claim first demonstration of this phenomenon. Our distinctness lies in cross-representational matching on a real behavioral surface plus explicit transfer and measurement audits. The full limitation inventory is in Appendix E.

7. Conclusion

In Gemma-3-4B-IT, strong internal readouts supported monitoring and sometimes supported control. But their usefulness as steering targets depended on the behavioral surface and the measurement procedure used to judge success. The four-stage audit framework (measurement, localization, control, and externality) summarizes that bounded lesson: passing one gate does not license claims about the next.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically evaluation methodology for safety-relevant model interventions. The wrong-entity substitution failure mode documented here has implications for deployment of steering-based safety mitigations: interventions that appear beneficial on constrained evaluation surfaces may corrupt factual generation in ways that are harder to detect than outright refusal.

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A. Detector Interpretation Audits

The top-weight CETT neuron (L20:N4288) is not stable across regularization settings: it is absent at $C \leq 0.3$, appears at $C = 1.0$, and drops to rank 5 at $C = 3.0$. The extreme weight is better explained by L1 concentration among correlated features than by unique causal importance. Additionally, full-response readouts are strongly length-confounded: response length dominates truthfulness signal by $\sim 3.7\times$ under mean aggregation. These are not threats to the intervention claims (FaithEval uses single-letter extraction), but they justify the narrower wording in the main text.

B. Construct Map

C. FaithEval Detailed Rates

D. Benchmark Power Summary

E. Limitation Inventory

Table 4. Benchmark construct map. Each evaluation surface measures a specific behavioral construct.

Benchmark	Construct	Primary Metric	Main Caution
FaithEval	Context-resistance under anti-compliance prompting	Compliance rate	Credulity lever, not standard truthfulness
TruthfulQA MC1/2	Answer selection under constrained candidates	MC1 accuracy	Does not measure open-ended generation
TriviaQA bridge	Short-form factual generation	Adj. accuracy	Single-rater failure coding
JailbreakBench	Harmful compliance under adversarial prompting	Graded harm rate	Binary eval. underpowered (MDE ∼6 pp)
BioASQ	Domain-specific factoid QA (biomedical)	Accuracy	Flat despite behavioral perturbation

Table 5. FaithEval compliance by method and scaling factor ($n =$ 1,000; Wilson 95% CIs).

α	Neurons	SAE H-feat.	SAE rand.
0.0	64.2%	72.3%	74.9%
0.5	65.4%	74.7%	74.8%
1.0	66.0%	66.0%	66.0%
1.5	67.0%	75.0%	75.0%
2.0	68.2%	75.1%	74.9%
2.5	69.5%	74.9%	74.9%
3.0	70.5%	69.9%	74.6%

Table 6. Benchmark power and minimum detectable effect (MDE) summary.

Benchmark	n	Slope	MDE
FaithEval	1,000	$+2.09$ pp/α	$\sim$3 pp
FalseQA	687	$+1.62$ pp/α	$\sim$4 pp
JailbreakBench	500	$+2.30$ pp/α	$\sim$5 pp
BioASQ	1,600	flat	$\sim$2 pp

Table 7. Key limitations and which claims they constrain.

#	Limitation
L1	Single model (Gemma-3-4B-IT); all claims.
L2	Matched-readout confound: SAE and neurons differ in operator form and layer coverage.
L3	SAE layer coverage is partial; a different width or layer set could yield non-null results.
L4	Bridge failure-mode coding is single-rater (no inter-rater reliability).
L5	Jailbreak H-neuron specificity is single-seed ($p = 0.013$); seeds 1–2 pending.